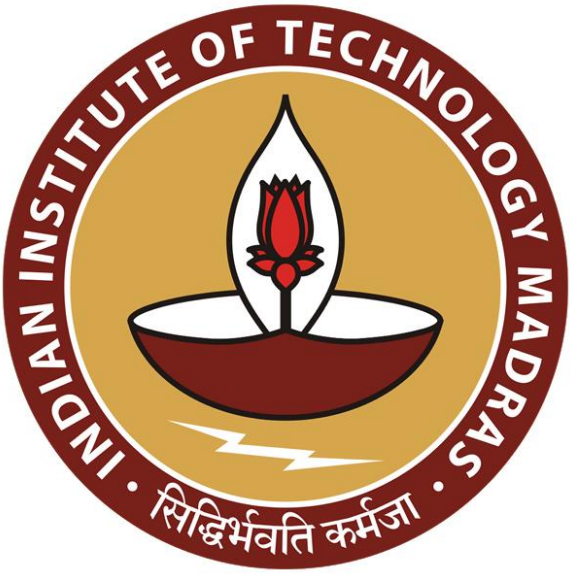
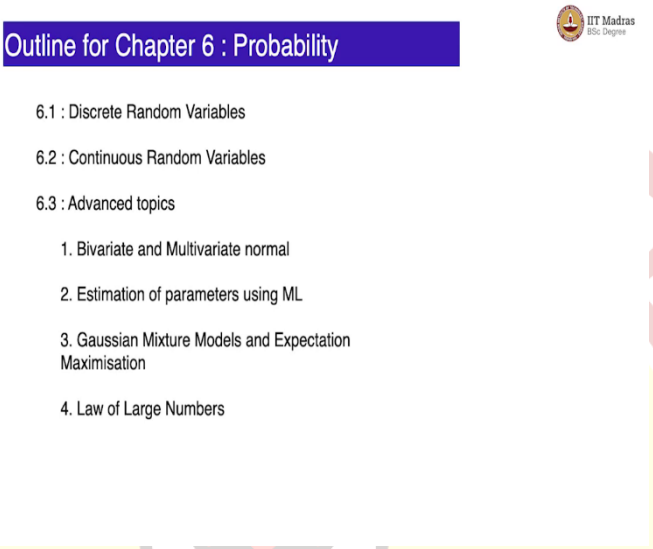


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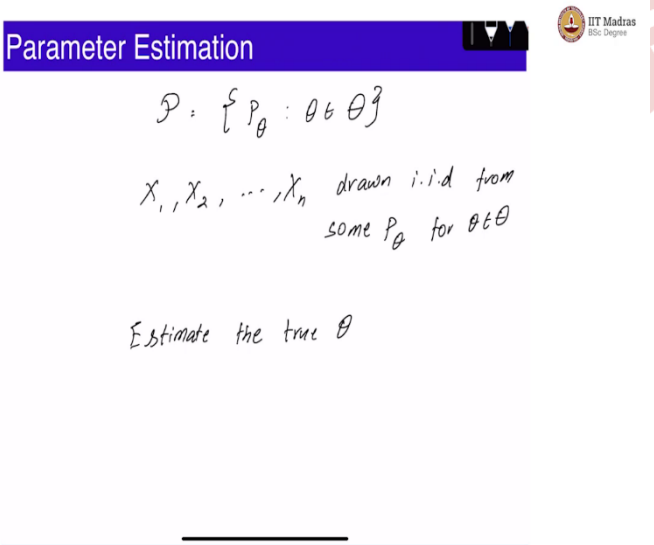
Machine Learning Foundations
Professor. Harish Guruprasad Ramaswamy
Department of Computer Science and Engineering
Indian Institute of Technology, Madras
Estimation of parameters using ML

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Hello everyone, and welcome to another lecture on probability. So, in this lecture, we will take a look at parameter estimation using mainly, a method known as maximum likelihood. This ML is not the same as machine learning. Maximum likelihood is older than machine learning, but maximum likelihood is one of the key tools used by machine learning so ML uses ML.

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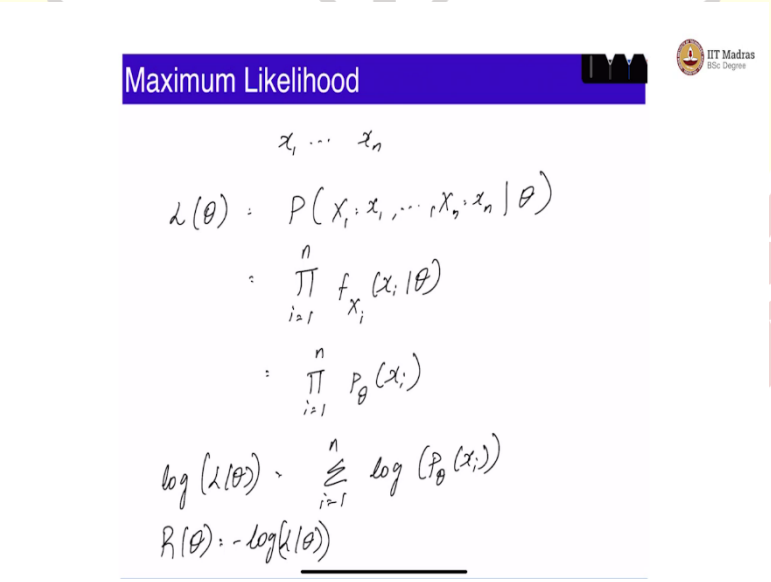


So, what is parameter estimation? So, here is the setting of parameter estimation is as follows. It is where you have a family of distributions. You do not have a single distribution, but rather a family of distributions, so that is typically denoted as script P and a family of distributions parameterized by theta. So, let us say P theta, and your theta takes values in some parameter space capital theta. So, what we have is a family of distributions.

So, we will see several examples of this pretty soon, what we have is you can just do it as a collection or a set of densities or distributions. So, you have such a object and you have data X1, X2, Xn drawn iid from some X1, X2 dot dot dot xn drawn iid from some P theta, for theta belonging to theta. You have your data drawn from some element in this family, but you do not know which.

The goal is to use this data and find out what is the best theta. So, use this to estimate the true theta. Theta represents the parameter, and from data can you estimate the true parameter theta? So, that is the parameter estimation set. How do you estimate the parameter of distribution from the, from some finite amount of data? So, there are several ways in which you can do this, but the most popular and the most prevalent method is what is called maximum likelihood.

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So, it is based on a very simple idea. So, you, if your theta was given, so you have data x1 to xn, so you can define what is called the likelihood of a parameter theta L of theta is simply equal to probability of X1 equal to X1 dot do dot Xn equal to Xn given theta. So now, if theta were given, then you can compute this. So, I am going to be using this notation for both

discrete and continuous even though for continuous this is strictly not correct. I should use the density, but I am going to use this meanwhile.

So, this effectively is saying that is fx I am going to use the same thing for probability mass function and density function. So, this is the idea. So, what is this capturing? This capturing for any given parameter theta, I can measure how likely is this data. So sorry for that. So, for any given parameter theta, how, what is the probability of this particular data set being generated? I can measure that.

And I will choose the parameter for which the probability of this data being generated is highest. That is the reason why it is called maximum likelihood. So, this is likelihood. This L of theta term called a likelihood and you want to choose the parameter theta, which maximizes the likelihood. So, let us do some simplification for the simplification.

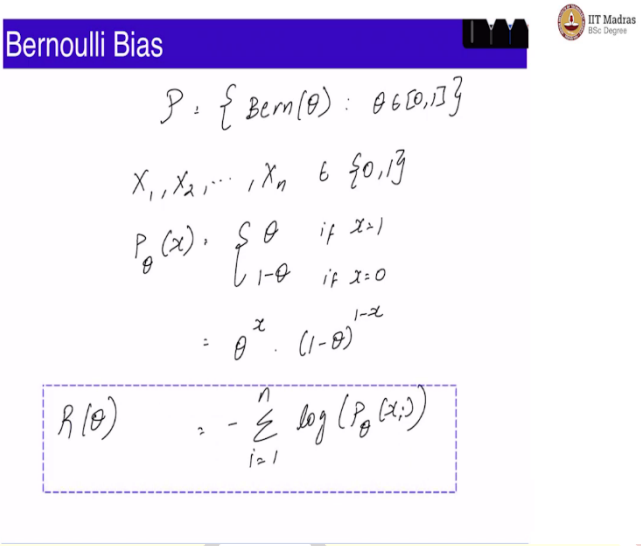
We have assumed that X1 to Xn are independent. So, this essentially boils down to product over i equal to 1 to n, fxi of xi given theta. So, theta is the parameter on fx, I explained it, and what is that, that is exactly equal to product or i equal to 1 to n P theta of Xi. Recall our family of distributions was parameterize by theta, so P theta of Xi. You want to maximize this, but you have a product term here, that is an ugly thing to maximize.

Is there an easy way you can do that? Well, you can take the log of this, instead of maximizing the likelihood you can maximize the log likelihood. If you did that log of L of theta is sum or equal to 1 to n log of P theta of xi. So, this is the log likelihood, maximizing the likelihood is the same as the maximizing the log likelihood because log is a monotone function.

Sometimes you might, instead of maximizing the log likelihood, you might minimize the negative log likelihood. Let us call the risk function R of theta to be the negative log likelihood, which is - log L of theta. So, you can easily see that maximizing the log likelihood is the same as minimizing the negative log likelihood, so that is, so it is just a issue of which you are comfortable with. Are you comfortable with maximizing our enemies, but they are really the same.

So, if you can find the theta, which maximizes this particular sum that is your estimate of the parameter. Is that the true parameter? We will never know. Because the moment you have probability, the moment you might, you can never be sure of this, but we can give the most likely parameter that generated this data.

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So, let us see some examples. Let us take the simplest example, which is the case of Bernoulli Bias. So, in this case, the parameter, the probability, the family of distributions P is simply Bernoulli with bias theta, where theta is in the range of 0 to 1. That is your family of distributions, this Bernoulli theta, and you have a sequence of let us say coin tosses, X1, X2, dot dot dot Xn all of these are either 0 or 1. Let us say they are heads or not heads, so it is 1 or 0.

From this, can you estimate the parameter theta. So, you can immediately see that you can never get the true theta. Even if you had 1 million coin tosses can you, you can measure I mean you can estimate the bias of the coin, but can you be certain, can you be 100 percent certainty that it was a fair coin. You can never be certain maybe it was 0.51, maybe it was 0.6.

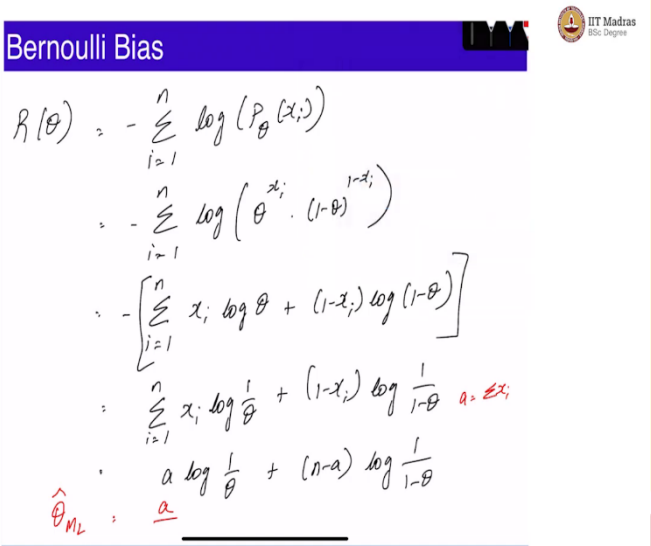
Just by randomness, it so happened that in a million coin tosses it happened in the way it does. It can never be 100 percent certain. The moment you are entering the realm of probability you can never be 100 percent certain anyway. So, all of this will be with respect to certain uncertainty, but you can still find the maximum likelihood parameter, which is a reasonable thing.

So, let us find the maximum likelihood parameter for this. And the answer you will find is a very natural solution, but we will try to get that in a dense way. So now first, what is P theta of x? P theta of x is equal to theta if x is equal to 1, and 1 - theta if x equal to 0, so heads with probability theta, and tails with probability 1 - theta. So, heads means 1 tails means 0.

This is fine, but we can write this in a more compact way as follows, which is theta power x into 1 - theta power 1 - x. So, note that this particular way of writing is exactly the same as this. So, x can take values only 1 or 0. If x equal to 1, this will be theta power 1 into 1 - theta power 0 so that is theta. If x was 0, this would be theta power 0 into 1 - theta, power 1, which is 1 - theta.

So, this case, expression is written in a compact way, as like this theta power x into 1 - theta power 1 - x. And this makes sense because x can take values will either 1 or 0. So, now we have P theta of x. Now let us write down the negative log likelihood which his R of theta which is a negative log likelihood, which is - sum over i equal to 1 to n log of P theta of Xi. Let us copy this and continue the next page. Sorry, I will have to create another page.

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Hope that another page is created. So, this is R of this is a negative log likelihood. And we already written P theta of xi, let us move this a little bit to the left. So, let us go ahead and simplify this further. This is - sum or i equal to 1 to n, log of theta power xi into 1 - theta power 1 - xi. And you can immediately see why we did this because log nicely decomposes products.

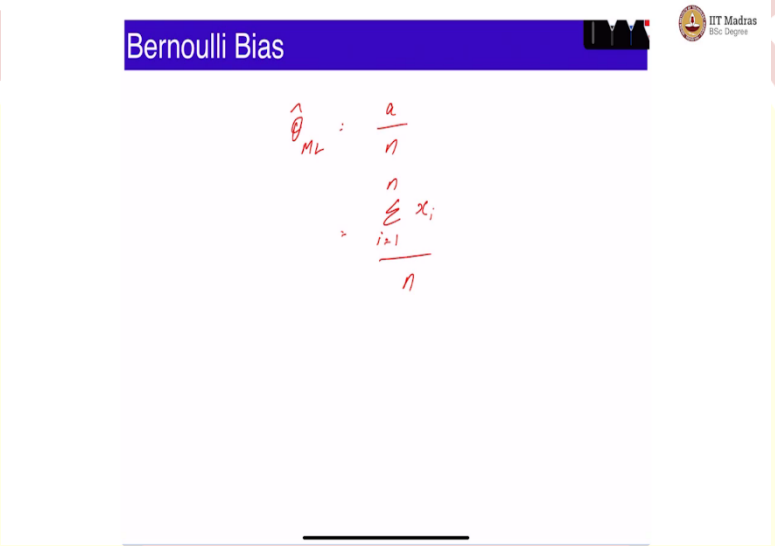
So, this can write as sum over i equal to 1 to n - sum over i equal to 1 to n, xi log theta + 1 - xi log of 1 - theta. This - is common across everything. If you can further simplify this as follows sum over i equal to 1 to n, xi log 1 by theta + 1 - xi log 1 by 1 - theta.

Let us simplify this even further as follows. So, the sum over i equal to 1 to n, that is called sum over i equal to 1 to n xi as the number of heads, let us call that a, like for simplicity, let

us call that a. So, it is a times log 1 by theta, and sum over i equal 1 to n, 1 - xi is simply the number of tails that is going to be equal to n - a + n - a log 1 by 1 - theta, where a is the number of heads, a is equal to sigma Xi the number of heads.

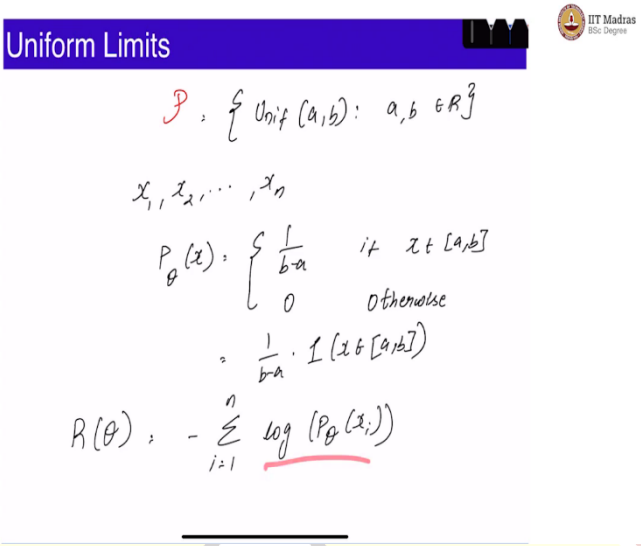
Now you want to find the theta which minimizes this. What theta minimizes this? This you might have seen in your optimization module, the theta that minimizes this is exactly given by it is a very natural thing. This theta, which minimizes this is the ML estimate the maximum likelihood estimate is theta hat ML is simply going to be equal to a by n. Apologies for that, theta hat ML simply going by a by n. I will just create another page here.

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Theta hat ML, which is the minimizer of the previous thing is simply given by a by n. This is a natural thing here, which is sum over i equal to 1 to n, xi by n. So, it is simply the fraction of the number of fraction of heads, that is your ML estimate, which is a natural thing to do. You might have already done this even without the ML estimate. But ML estimate gives you a formal way of saying that the most, well, the parameter, which is which most explains the data is given by simply the sampled mean or the number of the fraction of heads in the number of contests.

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So, let us do one more example where we have x, your P theta, your P a family of distribution P is simply given by uniform of a , b, as your a and b are reals. So, now there are two parameters here, not a single parameter, but rather two parameters. Your theta in this case would correspond to a vector if you want, it is a two-dimensional vector, theta corresponds to all the parameters in this case a and b.

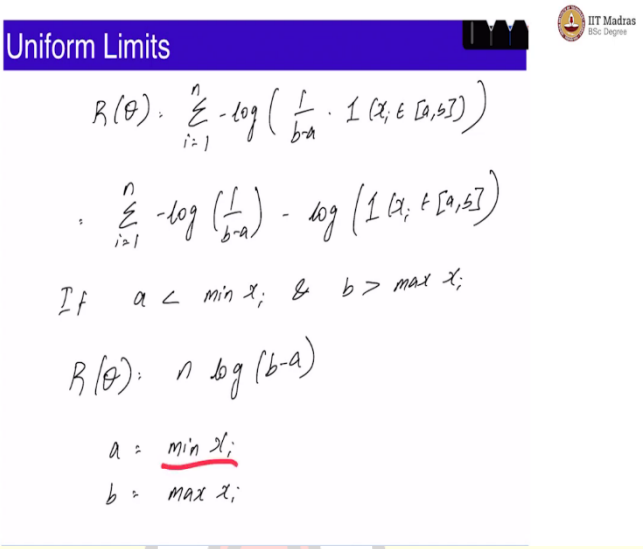
So now, you have uniform a and b and you have data your data b is simply given by let us say you have data X1, X2 dot, dot, dot Xn. So, what is the maximum likelihood estimate for a and b, given this data? You might already have an answer in mind. And you will see that the answer that you get at the end of this is also most likely matches your answer in mind, and it is a very intuitive answer.

Now let us what is P theta of x first? P theta of x is simply given by 1 by b - a theta, I am going to use theta as effectively a , b and so theta represents both a and b, so 1 by b - a if x belongs to a to b, and 0 otherwise. So, you have your P theta given as follows. Now, we can get the negative log likelihood function which is your, the risk of theta as simply - sum or equal to 1 to n log of P theta of Xi.

So, the same way we did the simplified P theta in a product form, we can do the same thing here and say P theta of x is simply 1 by b - a into the indicator of x belongs to a to b. This expression is exactly the, this expression is exactly the same as the previous expression, but just that this is more compact and expresses in a product form. And the product form

expression is nicer because log decomposed it even further. So, let us write down the risk of the parameter theta as follows.

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So, if you go ahead and write down your negative log likelihood, you want to minimize this function R of theta, as simply sum over i equal to 1 to n - log P theta of xi. So P theta of xi was simply xi belongs to 1 by b - a into indicator of xi belongs to a to b. Let us decompose this even further, this is equal to sum over i equal to 1 to n - log of 1 by b - a - log of indicator of xi belongs to a , b.

So, now we have R of theta expressed as follows. What can you say about this function? If you have your, even if one of the xis does not belong to a , b, this is going to be equal to 0 and log of 0 is going to be - infinity. So, sorry, - log of 0 is going to be log of 0 is going to be - infinity, - log of 0 is going to be + infinity.

So, you cannot have your xi you a, b such that even not even a single xi should be outside the limits. So, we can if however, now we have that if a is lesser than min over xi and b is greater than max over xi. So, if this is not true, then you have the R of theta is going to be equal to infinity. This is a finite term anyway, and you have, if this particular clause is not true, then R of theta is going to be infinity. So, you want to minimize the negative log likelihood, so you do not want to choose such a theta.

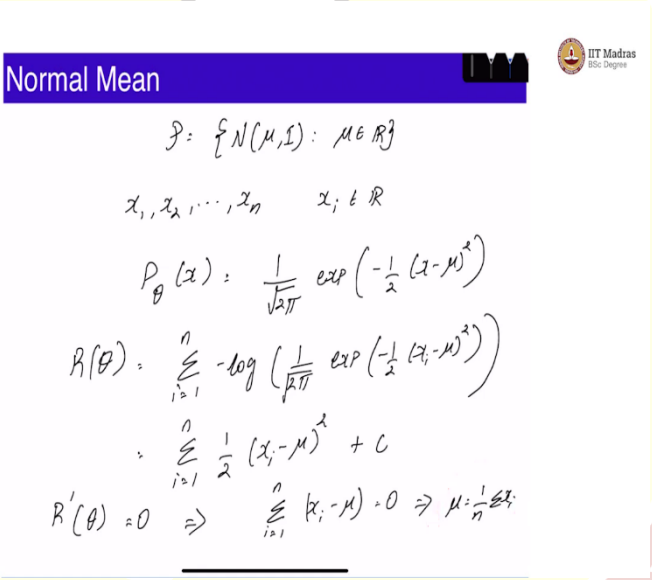
So, now, without loss of generality, we can choose a to b less than min of xi and b to be greater than max of xi. If you do choose that, then this term is always going to be the right, the second term is always going to be equal to 1. So, the log of 1 is going to be 0 it gets

completely canceled, so R of theta is simply n times log of b - a. So, you want to minimize this.

So, you have n times - log of 1 by b - a simply n times log of b - a. So, you want to minimize this term, you want to choose a and b which minimizes this. So, you want to minimize b - a, the only way you can do that is by setting a to be equal to min over xi and b to be equal to max over xi. So, this particular value of a and b is the maximum likelihood estimate of the data.

So, given a bunch of data points X1 to Xn the ML estimate for a is simply min over Xi with ML estimate for b simply max over Xi. This is a pretty natural estimate, but now we have derived this from those principles.

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Let us move on to another example, where we have normal mean. Probably the parameter, your family of distributions P is simply let us say normal with mean mu, and variance 1. The variance is a constant, variance is not a parameter. Your mu belongs to R. Now, this is your parameter theta is simply equal to mean mu. You have data points x1, x2 dot, dot, dot xn, each of these xis are real value.

What is the ML estimate of the parameter mu? What is P theta of x? Well, that is simply 1 by root 2 pi exp - half into x - mu the whole square. So, theta and mew I am going to use interchangeably because theta is the general parameter, mu is simply notation I am using for the case of normal random variable. So, P theta of x is simply 1 by root 2 pi x puff - half into x - mu the whole square. This is simply the density function of the normal random variable.

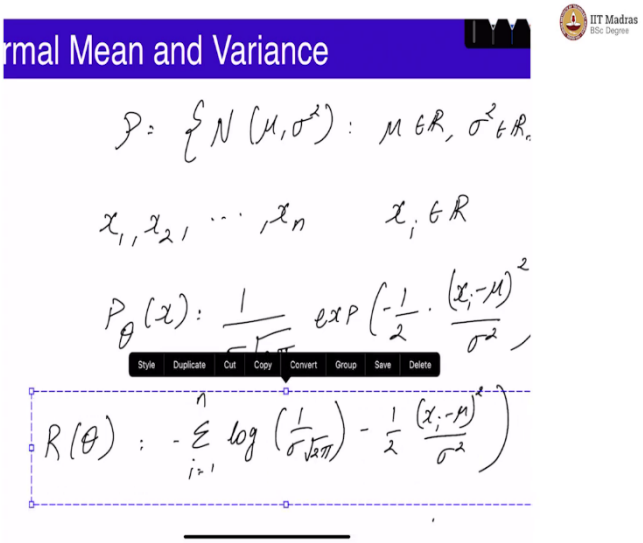
What is your risk of are the negative log likelihood of theta simply equal to sum over i equal to 1 to n - log of 1 by root 2 pi into exp - half into Xi - mu the whole squared. You can once again use the power of the log and you have log of 1 by root 2 pi, which is a constant, which does not depend on theta or theta in this case is simply mu. So, - log 1 by root 2 pi is something which does not depend on theta, we will just ignore that. We will just take terms which contain the parameter theta or mu so it is just.

Now, which means that we have sum over i equal to 1 to n, so you have - log here and x your log and x really cancel each other and you have the two - signs also cancel out each other, so you end up with R of theta is equal to sum over i equal to 1 to n half xi - mu the whole squared. This is your negative log likelihood function. You have some constants which you are, which do not depend on the parameter mu, we will just collect that together as a constant.

So, what value of mu minimizes this? Well, in this case, it looks like a function of mu. How do you find the mu which minimizes that? You can use your first start of necessary condition, you can take the derivative of this with respect mu and set to 0. And those are critical points. Your minima has to be one of these points. In this case, it happens to be that there is a unique point, so you can just solve for it and you get the minimum.

So, if you set your R dash of theta to be equal to 0, then you would get sum over i equal to 1 to n, xi - mu to be equal to 0, which implies mu to be equal to 1 by n sigma xi. This is also a very natural looking parameter that is you choose your ML estimate of the parameter mu is simply equal to the sampled mean, say 1 big n sigma xi. So that is your ML estimate for the normal distribution.

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Now let us do one more final example and wrap up. Your parameter family P is simply the set of normal distributions with mean mu and variance sigma squared. So, you have mu belongs to yield and sigma squared belongs to positive values. So, you are forced to choose positive values for the variance. We will actually have sigma squared, but sigma squared itself will use it is a variable. We will not try to find sigma but rather sigma square itself.

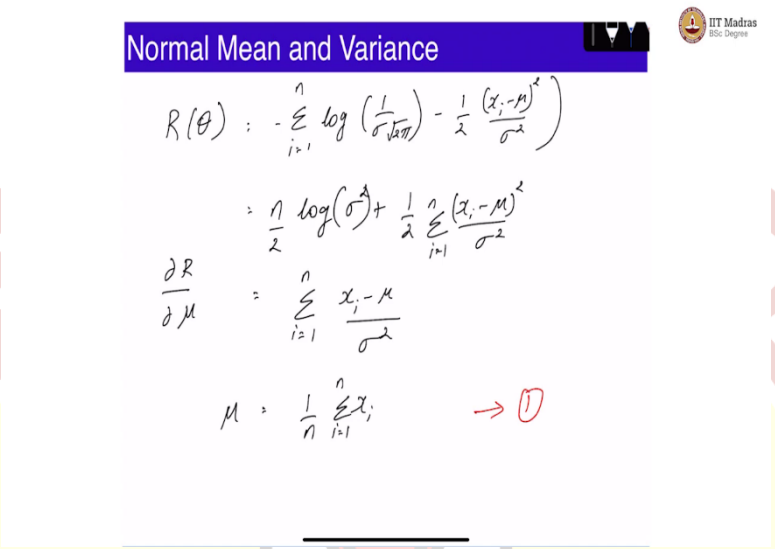
We want to find the ML estimates for the parameter mu and the parameter sigma squared. Once again what does the data going to look like? The data is going to look like x1, x2, dot, dot, dot xn. You have n data points each of these data points are real value. What is the ML estimate of what is first write on what is P theta of x? P theta of x is here you have the variance sigma is not 1, so you cannot ignore that, so we have 1 by sigma and that is also a parameter.

You have 1 by sigma, square root of 2 pi into x puff - 1 by 2 into Xi - mu, the whole squared by sigma squared. This is your P theta of x. Can we just use ML principle, the likelihood to get that right? So, what is the negative log likelihood R of theta your theta, the parameter here corresponds to the both mu and sigma squared. So, I am going to use theta as a package, which contains both mu and sigma squared. Let us we need couple more pages.

So, what is R of theta? R of theta is sum over i - sum over i equal to 1 to n, log of 1 by sigma square root of 2 pi +, - strictly speaking, - half into xi - mu, the whole squared by sigma squared. So, this - is common for both these terms. So, here in the when you had just the mean, you could just ignore this term. Why is that? Because this term happened to be just a

constant. But when you have sigma, although as a parameter, you cannot ignore this term, and you have to take care of this also. So, let us just copy paste this and continue in the next page.

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So, let us go ahead. But you can ignore the 1 by root 2 pi, because 1 by root 2 pi is a constant. So, you end up with n times log sigma, so you have 1 by sigma n - so cancels out get n log sigma + half sigma Xi - mu, the whole squared by the sum sigma and parameter sigma looks confusing, but I am going to just say that sum over i equal to 1 to n xi - mu the whole squared by sigma square.

So, this is your R of theta. Now you have two parameters, mu and sigma squared. Let us first take the, you can do the same thing again. You want to find the mu and sigma square, which minimizes this, so let us do the critical points. Sox let us take the derivative of this with respect to mu and sigma squared and set it to 0 and see if we can solve those two equations.

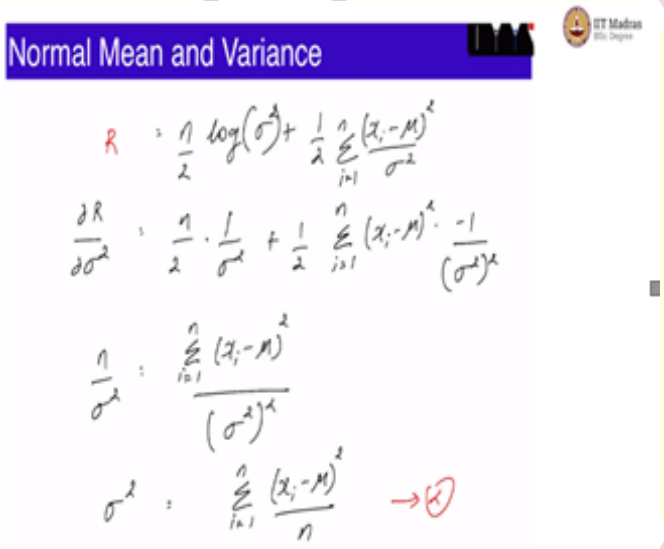
Let us first set the derivative with respect to mu and set it to 0. So, we have doh R by doh mu would be the first term that if you do that, the first time becomes 0, because that is not dependent on mu, so the second term would be sum over i equal to 1 to n xi - mu by sigma square. So, the half becomes half and 2 cancel out each other sigma squared stays here xi - mu, the whole squared becomes 2 into xi - mu so that two and a half cancel out. So, we have xi - mu by sigma squared.

So, which means that if you set this to be equal to 0, then the equation is mu would be equal to 1 by n sigma xi. This is something which you had seen already, you set your mean to be

equal to the sampled mean. This is the first equation for setting the gradient to 0 and finding the critical points. In this case, again, you will find that there is only one critical point and that does happen to be the minima of the negative log likelihood. So, now let us take the derivative of R with respect to sigma square and see where we go.

So, let us take the derivative with respect to sigma squared. So, if you do that we have a sigma here, it looks ugly. So instead of doing that, we can use n by 2 log, instead of having n log sigma, we will just call it n by 2 log sigma squared. So, this is simply the property of the log. We can do that instead. n by 2 log sigma squared + half sum over i equal to 1 to n xi - mu the whole squared by sigma square. Let us once again copy this for simplicity. And take the derivative with respect to sigma squared.

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So, if we take the derivative with respect to sigma square, doh R by doh sigma squared would be equal to n by 2 into 1 by sigma squared + half sum over i equal to 1 to n xi - mu the whole squared, so this is a constant as far as we are concerned. Now xi - mu the whole squared is a constant so xi - mu the whole squared into 1 by sigma squared 1 by sigma squared, if you take the derivative with respect to sigma squared it could be - 1 by sigma squared the whole square.

1 by x the derivative of this is - 1 by x squared, so - 1 by sigma squared the whole squared. So, now, if you take the derivative of with respect to sigma squared and set it to 0, you get 1 by n by sigma squared is equal to sum over i equal to 1 to n xi - mu the whole squared by sigma squared the whole square. So, one of the sigma squared terms cancels out you get

sigma squared simply equal to sum over i equal to 1to n xi - u the whole squared by n. This is your second equation.

You can solve the first and second equation together, and you get your mu to be equal to the sampled mean and your sigma squared to be equal to the sampled variance computed using mu the empirical mean, as the empirical mean, mu is equal to the sampled mean, and sigma squared is equal to the sampled variance.

So, those happen to be the ML estimate for mu 1 sigma squared for the case of the normal family with mean and variance, as the parameters. So, ML, we have seen some very simple examples of maximum likelihood, and these found the building blocks. But you can see, even for very complicated models, the maximum likelihood forms the first point from which you build on more complicated algorithms.

We will see such an example in the next class where we will define a pretty complicated model on as the Gaussian mixture model. And how the parameters estimation for that reduces to, is very similar to a very fundamental machine learning problem called the clustering and how we can apply the maximum likelihood method for solving that particular estimation problem. With that we can wrap up this class.